Simetría salvo una derivada total

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$\mathcal{L} = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) - mgy$ (I) $x \to x+s$ (II) $x \to x+st$ (III) $y \to y+st$

a) $\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial q_j}\delta q_j + \frac{\partial\mathcal{L}}{\partial \dot{q}_j}\delta \dot{q}_j$

$\delta\mathcal{L} = \frac{df}{dt}$ $f(q_j, t)$ entonces la transformación es una simetría de Noether y hay una cantidad conservada.

(I) $\delta x = s$ $\delta\dot{x} = 0$

$\frac{\partial\mathcal{L}}{\partial x} = 0$ $\frac{\partial\mathcal{L}}{\partial \dot{x}} = m\dot{x}$

$\delta\mathcal{L} = 0(s) + m\dot{x}(0)$ $\delta\mathcal{L} = 0 = \frac{df}{dt}$ $f(x,t) = 0$

(II) $x \to x+st$

$\delta x = st$ $\delta\dot{x} = s$

$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial q_j}\delta q_j + \frac{\partial\mathcal{L}}{\partial \dot{q}_j}\delta \dot{q}_j$

$\delta\mathcal{L} = 0(st) + m\dot{x}(s) = m\dot{x}s$ $f(x,t) = mxs$

(III) $y \to y+st$ $\delta y = st$ $\delta\dot{y} = s$

$\frac{\partial\mathcal{L}}{\partial y} = -mg$ $\frac{\partial\mathcal{L}}{\partial \dot{y}} = m\dot{y}$

$\delta\mathcal{L} = -mg(st) + m\dot{y}(s)$

$\delta\mathcal{L} = s(m\dot{y} - mgt)$

$\frac{d}{dt}$

$\delta\mathcal{L} = \frac{1}{s}\frac{d}{dt}\left(my - \frac{1}{2}mgt^2\right)$

$smgt$ $\frac{d}{dt} = \left(\frac{1}{2}mgt^2\right)$

$\frac{d}{dt}(my) = m\dot{y}$

→ no existe

b) $J = \sum_{j=1}^{s}\frac{\partial\mathcal{L}}{\partial \dot{q}_j}\delta q_j - f = cte$

(I)

$J = m\dot{x}(s)$

(II)

$J = m\dot{x}(st) - mxs$

$J = ms(\dot{x}t - x)$

(III)

No existe corriente de Noether

c) (I)

$p_x = m\dot{x}$

$p_x(t) = m\dot{x}(t) = m(V_0\cos\theta) = cst$

(II)

$Q_2 = t\dot{x} - x$

$Q_2(t) = t\cdot(V_0\cos\theta) - (V_0\cos\theta\cdot t$

$= V_0\cos\theta\cdot t - V_0\cos\theta\cdot t$

$= 0$

(III)

$J_3 = m\dot{y}\cdot st - \left(smyt - \frac{1}{2}smgt^2\right) = sm(t\dot{y} - y) + \frac{1}{2}smgt$

$t(V_0\sin\theta - gt) - h - V_0\operatorname{sen}\theta\cdot t + \frac{1}{2}gt^2$

$= tV_0\operatorname{sen}\theta - gt^2 - h - V_0\operatorname{sen}\theta\cdot t + \frac{1}{2}gt^2$

$= -gt^2 - h + \frac{1}{2}gt^2$

$= -h - \frac{1}{2}gt^2$

$J(t) = sm\left(-h - \frac{1}{2}gt^2\right) + \frac{1}{2}smgt^2$

$= -smh - \frac{1}{2}smgt^2 + \frac{1}{2}smgt^2$

$= -smh$

$\frac{dJ}{dt} = \frac{d}{dt}(-smh) = 0$

d)

(I) La cantidad conservada es p_x, como no hay fuerzas en la dirección x, la aceleración en x es cero: $\ddot{x} = 0$ y el momentum $p_x = m\dot{x}$ se conserva en todo momento

(II)

La cantidad $Q_2 = t\dot{x} - x = t\cdot V_0(\cos\theta - V_0\cos\theta\cdot t = 0$

(III)

La existencia de una fuerza externa quiebra simetrías